



Vinod Suresh Badni
Bachelor of Engineering
Computer Science Engineering
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Portfolio: Vinod's Portfolio

Professional Summary

Computer Science graduate with strong skills in Python, C, C++, and machine learning. Completed a research internship focused on the generation of GAN-based textile patterns. Proficient in Software Development, DSA, and AI/ML tools such as TensorFlow and PyTorch. A quick learner with a passion for building scalable, real-world tech solutions.

Experience

- University Of Boras Feb-Jul 2025
AI research Intern Remote
 - Led research on automated textile pattern generation using advanced Generative Adversarial Networks (GANs), addressing challenges in design scalability and structural coherence.
 - Designed and implemented an enhanced Wasserstein GAN with Gradient Penalty (WGAN-GP), incorporating self-attention and residual blocks, outperforming baseline WGAN.
 - Achieved a 23.1% reduction in Fréchet Inception Distance (FID) (from 36.8 to 28.3), alongside improvements in Inception Score (IS) (3.4 to 4.8) and Structural Similarity Index (SSIM) (0.58 to 0.75) on a 17,000-image textile dataset.
 - Conducted comprehensive quantitative and qualitative analysis, including loss curve evaluation and visual sample assessment, to validate model performance and stability.
- Agnirva AI Internship Program May - Aug 2025
AI Internship KLETECH Hubli
 - Selected from a nationwide pool for a prestigious AI internship focusing on real-world applications of Artificial Intelligence and Machine Learning.
 - Participated in expert-led sessions covering advanced topics such as Generative AI, Large Language Models (LLMs), Computer Vision, NLP, and Responsible AI.

Technical Skills

Programming Languages: C, C++, Python.
Technical Skills: Data Structures & Algorithms, Object Oriented Programming, Operating Systems.
AI/ML Libraries: TensorFlow, Pytorch, Keras.
Database: SQL.
Web Technologies: HTML5, CSS, Reactjs.
Areas of Interest: Software Engineer, AI/ML Engineer, Data Analyst.

Projects

- SAR Image Colorization for Comprehensive Insight Using Deep Learning Model
Tools & Technologies used: Python, PyTorch, Pix2Pix GAN, Machine Learning.
 - * Developed a deep learning model using Pix2Pix GAN to colorize grayscale SAR images.
 - * Implemented a generator and discriminator architecture to train the model on paired SAR and optical images.
 - * Optimized the model's loss functions to improve the accuracy and visual quality of the colorized images.
- E-commerce Database Management System
Tools & Technologies used: SQL.
 - * Developed Entity-Relationship (ER) diagrams, identified entities, attributes, and relationships with cardinality ratios for a comprehensive e-commerce system.
 - * Normalized the database schema to BCNF, optimizing data integrity and reducing redundancy across relations like Customer, Products, Shopping Cart, and Payment.
 - * Executed complex SQL queries involving joins, subqueries, and views to extract and analyze e-commerce data.
- Car Pooling Management System
Tools & Technologies used: ReactJS, JAVA, MongoDB.
 - * Developed a web application to facilitate ride sharing between drivers and riders with similar routes or destinations.
 - * Designed and implemented key modules for driver and administrator, enabling seamless registration, matching, and ride management.

- * Focused on reducing traffic congestion, carbon emissions, and transportation costs, aligning with UN SDG 11: Sustainable Cities and Communities.
- Shortest path between stations of Hyderabad Metro station using DSA
- Tools & Technologies used: C, DSA.
- * Calculated the shortest distance and cost between a single source station and a destination station using Dijkstra’s algorithm.
 - * Designed and implemented a console-based application to calculate the shortest path, travel time, and cost between metro stations in Hyderabad.
 - * Calculated distances from a single source station to all other stations and single source to single destination.

Education

*KLE Technological University2021-2025

Bachelor of Engineering in Computer Science

Certifications / Achievements

*Python Fundamentals for Beginners.Great Learning, 2022

*TATA Group- GenAI Powered Data Analytics Job SimulationTATA, 2025

*Participated in Smart India HackathonISRO, 2024